AIML INTERNSHIP — WEEK 1

AI Concepts Report

Student Name	Shaik Arshiya Tabasum
Roll Number	24JR1A0529
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Q1. What is Artificial Intelligence?

Artificial Intelligence (AI) is the branch of computer science that focuses on creating machines and software capable of performing tasks that typically require human intelligence. These tasks include understanding natural language, recognising images, making decisions, solving complex problems, and learning from experience.

AI systems are built using algorithms and large amounts of data. They can be trained to identify patterns and make predictions without being explicitly programmed for every scenario. Today, AI is used in a wide range of applications including virtual assistants (like Siri and Alexa), recommendation engines (Netflix, YouTube), self-driving cars, medical diagnosis, and fraud detection.

Key Characteristics of AI:

- Learning — AI systems improve their performance over time using data.
- Reasoning — AI can analyse situations and draw logical conclusions.
- Problem Solving — AI can find solutions to complex, multi-step problems.
- Perception — AI can interpret inputs like images, audio, and text.
- Adaptability — AI adjusts its behaviour based on new information.

Q2. Difference between AI, Machine Learning, Generative AI, and Agentic AI

These four terms are related but represent different levels and types of artificial intelligence:

Term	Definition	Example
Artificial Intelligence (AI)	The broadest field — any technique that enables machines to mimic human intelligence.	Chess engines, spam filters, voice assistants
Machine Learning (ML)	A subset of AI where systems learn from data and improve automatically without explicit programming.	Email spam detection, Netflix recommendations
Generative AI	A type of AI that creates new content — text, images, music, code — based on patterns from training data.	ChatGPT, DALL-E, Claude, Stable Diffusion
Agentic AI	Advanced AI that autonomously plans, makes decisions, uses tools, and completes multi-step tasks with minimal human input.	AutoGPT, AI coding agents, AI research assistants

Q3. Three Open-Source AI Models on Hugging Face

Hugging Face is an AI platform that hosts thousands of open-source AI models. Three notable models available there are:

1. BERT (Bidirectional Encoder Representations from Transformers)

Developed by Google, BERT is a powerful natural language understanding model. It reads text in both directions simultaneously, giving it a deep understanding of word context. BERT is widely used for tasks like question answering, text classification, and named entity recognition. Model ID on Hugging Face: bert-base-uncased

2. GPT-2 (Generative Pre-trained Transformer 2)

Developed by OpenAI and available on Hugging Face, GPT-2 is a text generation model. It generates coherent and contextually relevant text given a starting prompt. It is used for creative writing, autocomplete systems, and chatbots. Model ID on Hugging Face: gpt2

3. Whisper (by OpenAI)

Whisper is an automatic speech recognition (ASR) model developed by OpenAI. It can transcribe spoken audio into text in multiple languages with high accuracy. It is used for subtitles, voice-to-text applications, and accessibility tools. Model ID on Hugging Face: openai/whisper-base

Q4. Traditional Programming vs AI-Assisted Development

Below is a comparison between traditional programming and AI-assisted development:

Aspect	Traditional Programming	AI-Assisted Development
Approach	Developer writes explicit rules and logic for every scenario.	AI learns patterns from data and generates logic automatically.
Flexibility	Rigid — only handles cases the programmer anticipated.	Flexible — adapts to new situations it was not explicitly programmed for.
Data Dependency	Does not require large datasets.	Requires large, quality datasets for training.
Maintenance	Manual updates needed for every rule change.	Model retraining updates the system behaviour.
Example	A spam filter using if-else keyword rules.	A spam filter trained on millions of real emails using ML.
Best For	Predictable, well-defined, rule-based tasks.	Complex, dynamic tasks with uncertain or evolving inputs.

Q5. AI Hallucinations

AI hallucination refers to when an AI model generates information that is confidently stated but is factually incorrect, fabricated, or nonsensical. This happens because AI models like LLMs predict the next most likely word based on patterns in training data — they do not actually know facts or verify information before responding.

Why It Happens:

- The model has gaps or biases in its training data.
- It prioritises fluency and confidence over factual accuracy.
- It has no built-in mechanism to verify real-world facts.

Example of AI Hallucination:

If you ask an AI: 'Who is the CEO of XYZ Tech Company as of today?' — the AI might confidently give a name that sounds plausible but is completely fabricated, especially if the company is small and not well-represented in its training data. The AI does not know it is wrong — it simply generates the most statistically likely-sounding response.

How to Avoid Being Misled:

- Always verify AI-generated facts from reliable sources.
- Use AI tools with web search integration for factual queries.
- Treat AI output as a starting point, not a final source of truth.